UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	PROGRAMMING I
COURSE CODE	:	CSC402
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC402 mock paper (November 2026). Generated answers show working; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 10 x 1 mark; Part B = 5 questions (60 marks); Part C = 2 x 15 marks; total = 100 marks.

4. Questions: 17. Slots: 22. Placeholders: 0. No LLM call. Questions and answers are for local study use only.

5. Structure derived from 21 unique CSC402 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

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PART A — ANSWER KEY

Q	Ans	Question	Solution
A1	A	What happens when an array index is outside its valid range?	C++ does not perform automatic bounds checking.
A2	C	Which operator reads data from cin?	The extraction operator >> reads from an input stream.
A3	A	What does a compiler do?	A compiler translates the complete source file into object code.
A4	B	Which values can a bool variable hold?	A bool holds true or false.
A5	A	Which declaration creates a floating-point variable named price?	float is used for real numbers.
A6	C	Which statement is most suitable for selecting among several constant cases?	switch selects a case based on a constant expression.
A7	D	Which data type stores a whole number?	int stores whole numbers; float stores real values.
A8	D	Which statement immediately exits a loop?	break terminates the nearest loop or switch.
A9	D	How many times does a while loop execute when its condition is false initially?	while is a pretest loop; a false condition skips the body.
A10	C	Which term describes a finite sequence of precise steps that solves a problem?	An algorithm is a finite, ordered set of unambiguous steps.

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PART B — WORKED ANSWERS

QUESTION 1

Topic: debugging testing

a) Define algorithm and pseudocode.

An algorithm is a finite sequence of precise steps that solves a problem. Pseudocode is a structured English-like description of an algorithm.

Marking: Award 1 mark for each correct definition.

(2 marks)

b) Explain THREE types of programming errors and give one example of each.

Syntax error: missing semicolon. Runtime error: division by zero. Logic error: using < instead of <= in a condition.

Marking: Award 0.5 mark for each type and 0.5 mark for a valid example.

(3 marks)

QUESTION 2

Topic: fundamentals

a) Draw a flowchart for the grade classification.

Start -> read score -> decision score >= 80 -> A; else score >= 60 -> B; else score >= 50 -> C; else score >= 40 -> D; else F -> display grade -> end.

Marking: Award marks for terminal symbols, input, four/five decisions, each output and correct flow.

(8 marks)

b) Write pseudocode for the algorithm and convert it into a complete C++ program.

IF score >= 80 THEN grade = A ELSE IF score >= 60 THEN grade = B ... ENDIF; use if/else if in C++ and display grade.

Marking: Award marks for correct pseudocode, complete C++ structure, conditions, input and output.

(5 marks)

QUESTION 3

Topic: data types

a) Explain the difference between implicit and explicit type conversion, and give one C++ example of each.

Implicit conversion is automatic (int to double); explicit conversion uses a cast such as double(total) or static_cast<double>(total).

Marking: Award marks for both definitions and valid examples.

(6 marks)

b) Write a C++ program segment that calculates the average of three integer marks as a double and displays it to two decimal places.

double average = (m1 + m2 + m3) / 3.0; cout << fixed << setprecision(2) << average;

Marking: Award marks for floating-point division, fixed/setprecision and output.

(5 marks)

QUESTION 4

Topic: input output

a) Trace the program and state the output.

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Marking: The loop adds 1+2+3+4+5. Award marks for the sequence and final output.

(6 marks)

b) Rewrite the loop using a while loop. Then explain what changes if the condition is i < 5.

int i = 1; while (i <= 5) { total += i; i++; } With i < 5 the result is 10 because 5 is not added.

Marking: Award marks for correct while structure, update and changed result.

(9 marks)

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QUESTION 5

Topic: arrays

a) Declare two parallel arrays for five sports and initialise them with suitable values.

string sports[5] = {"Swimming","Badminton","Athletics","Cycling","Hockey"}; int medals[5] = {8,12,7,5,9};

Marking: Award marks for both declarations, correct sizes and related indices.

(6 marks)

b) Write a program segment to display each sport with its medals, then calculate the total and average medals.

Loop i=0..4 to display sports[i] and medals[i], accumulate total, average = double(total)/5.

Marking: Award marks for paired display, total, average and output.

(10 marks)

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PART C — WORKED ANSWERS

QUESTION 1

A cafe sells three items: coffee RM8.50, tea RM6.00 and juice RM9.50. A 10% discount is given when the subtotal is at least RM50.

Write a complete C++ program that reads the item code and quantity, calculates the subtotal and discount, and displays a formatted receipt with the item name, quantity, subtotal, discount and amount to pay.

```
#include <iostream>
#include <iomanip>
#include <string>
using namespace std;
int main() {
    int code, quantity;
    double price = 0, subtotal, discount = 0;
    string name;
    cout << "Item code (1 coffee, 2 tea, 3 juice): ";
    cin >> code;
    cout << "Quantity: ";
    cin >> quantity;
    if (code == 1) { name = "Coffee"; price = 8.50; }
    else if (code == 2) { name = "Tea"; price = 6.00; }
    else if (code == 3) { name = "Juice"; price = 9.50; }
    else { cout << "Invalid item code" << endl; return 1; }
    subtotal = price * quantity;
    if (subtotal >= 50) discount = subtotal * 0.10;
    cout << fixed << setprecision(2);
    cout << "Item: " << name << endl;
    cout << "Quantity: " << quantity << endl;
    cout << "Subtotal: RM" << subtotal << endl;
    cout << "Discount: RM" << discount << endl;
    cout << "Amount to pay: RM" << subtotal - discount << endl;
    return 0;
}
```

Marking: Award 3 marks for input, 5 for price/subtotal, 4 for discount, 2 for formatting and 1 for completeness.

(15 marks)

QUESTION 2

A bookstore records up to five books. Each book has a title, price and quantity in stock.

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

```
#include <iostream>
#include <string>
using namespace std;
struct Book {
    string title;
    double price;
    int quantity;
};
int main() {
    Book books[5];
    double totalValue = 0;
    int mostExpensive = 0;
    for (int i = 0; i < 5; i++) {
        cin.ignore();
        cout << "Title: ";
        getline(cin, books[i].title);
        cout << "Price and quantity: ";
        cin >> books[i].price >> books[i].quantity;
        totalValue += books[i].price * books[i].quantity;
        if (books[i].price > books[mostExpensive].price) {
            mostExpensive = i;
        }
    }
```

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```
    cout << "Total inventory value: RM" << totalValue << endl;
    cout << "Most expensive: " << books[mostExpensive].title
         << " (RM" << books[mostExpensive].price << ")" << endl;
    return 0;
}
```

Marking: Award 3 marks for struct definition, 4 for input, 4 for inventory value, 3 for most expensive and 1 for output.

(15 marks)

END OF MARKING SCHEME

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